

Credit Portfolio Risk Analysis

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Portfolios: Portfolio 1,2 & 3

Course: Introduction to Credit Risk and Applications in Python

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1. Introduction

At the core of credit risk management lies the modeling of dependencies among obligors. Default events are not independent; they are driven by common factors that cause defaults to cluster during periods of economic stress. The academic literature has developed structurally distinct model classes to capture this dependence. The present analysis employs both a latent-variable model in the tradition of Merton (1974) and a Bernoulli mixture model. Both frameworks are implemented for three real-world corporate portfolios and their results are compared systematically.

Here we assess credit risk using the risk measures Expected Loss (EL), Value-at-Risk (VaR), and Expected Shortfall (ES). Each risk measure is computed both analytically and by Monte Carlo simulation, allowing a direct comparison of the two evaluation approaches.

2. Data

The analysis is based on seven raw datasets provided for the assignment, which collectively cover three distinct credit portfolios. Each portfolio file contains an entity code that uniquely identifies the obligor, together with the corresponding Exposure at Default (EAD). The master data table (`entities_info`) records, for each entity, the company name, industry classification, region, Loss Given Default (LGD), and an estimated probability of default (`base_pd_hint`). Historical default events are provided in a separate dataset (`default_history_20y`) spanning the annual cohorts from 2006 through 2025. In addition, daily return histories are available at both the individual-entity level (`stock_returns`) and the industry-index level (`industry_index_returns`) for the period January 2, 2023, to December 31, 2025. Although the individual stock returns are not employed in the chosen modeling approach, the industry index returns serve as the basis for estimating the inter-factor correlation in the two-factor extension for Portfolio 3 (Section 3.2.1). In principle, the individual stock returns could also be used to derive asset correlations, an approach advocated by Düllmann, Küll, and Kunisch (2010).

The raw datasets exhibit a range of inconsistencies that, if left uncorrected, would lead to erroneous joins across tables and ultimately to biased model outputs. The following material issues were identified during an exploratory data analysis:

- **Entity_code:** Multiple datasets contain entity codes with inconsistent capitalization, leading whitespace, and—most critically—missing hyphens. This formatting discrepancy causes silent join failures and constitutes the single most consequential data quality issue encountered in the analysis.
- **Industry_code:** Eight distinct spellings for only two underlying industries (“Industry A”, “industry a”, “Ind. A”, “IndustryA”, and analogous variants for Industry B).
- **Missing values.** Six LGD values and five PD values are missing in `entities_info`. The `default_history` dataset contains 47 missing `default_event` entries. Across the three portfolios, nine EAD values are absent.
- **Duplicates.** Duplicate records in `entities_info` (3), `default_history` (15 entity-year pairs), `stock_returns` (20 entity-date pairs), and each portfolio (one per file). All duplicates are value-consistent.
- **Portfolios.** Beyond the EAD and entity-code issues noted above, each portfolio contains one duplicate entry with identical values.

The code addresses several additional minor issues (e.g., region spelling normalization, date format inconsistencies, removal of non-model columns). Since these do not materially affect the model outputs, the reader is referred to the code documentation for details.

The cleaning procedure follows a documented pipeline in the accompanying Python code. The key decisions and their rationales are summarized below.

- All entity codes are standardized via `strip()` and `upper()`. Additionally, codes lacking a hyphen are corrected by inserting one after the first character.
- The eight industry label variants are mapped to two canonical values using an explicit dictionary, ensuring that each entity is unambiguously assigned to one of the two industries.
- Duplicates across all datasets are removed after verifying that all duplicate records are value-consistent across all columns.
- Six missing LGD and five missing PD values are imputed using industry-specific means. While the overall distributions exhibit moderate dispersion, the within-industry distributions are narrow, rendering industry-mean imputation preferable to exclusion of the affected records.
- Nine positions without EAD are excluded. Since the EAD serves as the fundamental weighting variable in the loss formula, an imputed value would impose arbitrary assumptions on the portfolio composition. Exclusion is therefore the methodologically defensible choice.
- The 47 missing `default_event` entries are excluded rather than imputed. Since defaults are rare events (approximately 3% of observations), the default correlation estimate is highly sensitive to even small distortions in the observed default rate. Treating all missing values as non-defaults would artificially inflate the number of observed non-default years without empirical justification.

After all cleaning steps, three portfolios remain:

- Portfolio 1 with 28 positions and a total EAD of 631.1 million;
- Portfolio 2 with 28 positions and a total EAD of 704.5 million; and
- Portfolio 3 with 35 positions and a total EAD of 769.8 million.

Portfolios 1 and 2 are industry-pure: portfolio 1 contains exclusively industry A obligors, while portfolio 2 consists entirely of industry B obligors. Portfolio 3, by contrast, is a mixed portfolio with 26.3% of EAD attributable to industry A and 73.7% to industry B. The EAD-weighted mean probabilities of default are $p = 1.99\%$ for portfolio 1, $p = 3.52\%$ for portfolio 2, and $p = 3.12\%$ for portfolio 3. The EAD-weighted mean LGDs lie in a narrow band between 45.8% and 49.2% across all three portfolios.

3. Methodology

Two credit risk models are employed in the analysis of the three portfolios: a Beta-Binomial mixture model and a Gaussian one-factor threshold model.

The Beta-Binomial model (BB) belongs to the class of Bernoulli mixture models and draws on the idea, introduced by Joe (1997), of using a Beta distribution as the mixing distribution. Frey and McNeil (2002) developed this approach further into a practical framework for portfolio

credit risk; their subsequent work (Frey and McNeil 2003) identifies it as one of the canonical mixture specifications.

The Gaussian one-factor model (also referred to as the threshold model or the Vasicek model) belongs to the class of latent variable models and was introduced in its modern form by Vasicek (2002) as a specification for the loss distribution of a credit portfolio.

For both models, the following one year risk measures are computed: Expected Loss (EL), Value-at-Risk (VaR) at the 99% and 99.9% confidence levels, and Expected Shortfall (ES) at the same levels. VaR and ES are evaluated both analytically and by Monte Carlo simulation with $N = 100,000$ scenarios.

A key methodological decision concerns the EAD-weighting of the portfolio parameters p and LGD. Since each obligor contributes to the portfolio loss in proportion to its exposure, the EAD-weighted average ensures internal consistency between the calibration of p and the loss formula.

3.1 Beta-Binomial Model

The model specification follows Frey and McNeil (2003, Example 4.3). The key assumption is that the common systematic default probability Q in each period is a Beta-distributed random variable:

$$Q \sim \text{Beta}(\alpha, \beta).$$

All m obligors are jointly affected by the realization of Q , which generates the dependence structure. Conditional on Q , the default indicators Y_i are independent Bernoulli variables, so that the number of defaults M follows a Beta-Binomial distribution:

$$M | Q \sim \text{Binomial}(m, Q),$$

and unconditionally

$$M \sim \text{Beta-Binomial}(m, \alpha, \beta).$$

The Beta-distribution input parameters α and β are derived from the unconditional default probability p and the default correlation ρ_Y . The default correlation between any two obligors is related to the variance of the mixing distribution as $\rho_Y = \text{Var}(Q) / [p(1-p)] = 1/(1+s)$, where $s = \alpha + \beta$ is the concentration parameter of the Beta distribution. This Method-of-Moments (MM) estimator is described by Düllmann, Küll, and Kunisch (2010), who trace it back to Gordy (2000). The MM approach is preferred over a direct empirical computation of default correlations, since the portfolios under analysis contain few defaults, rendering a direct estimate highly unstable.

Under the homogeneity assumption, the number of defaults M is translated into a portfolio loss rate via $L = (M/m) \cdot \text{LGD_mean}$. The analytical risk measures are then computed directly from the closed-form probability mass function of the Beta-Binomial distribution.

3.2 Gaussian One-Factor Model

The Gaussian one-factor model, introduced by Vasicek (2002), assigns to each obligor a standardized latent asset variable:

$$X_i = \rho \cdot F + \sqrt{1 - \rho^2} \cdot \varepsilon_i,$$

where $F \sim N(0,1)$ is the common systematic risk factor, $\varepsilon_i \sim N(0,1)$ is the idiosyncratic component (independent of F and all other ε_j), and ρ is the factor loading.

A default occurs whenever $X_i \leq d = \Phi^{-1}(p)$. The factor loading ρ governs the relative importance of systematic versus idiosyncratic risk; the asset correlation between any two obligors is:

$$\text{Corr}(X_i, X_j) = \rho^2.$$

The asset correlation ρ^2 is derived from the default correlation ρ_Y (already estimated for the BB model) via the identity $\rho_Y = [\Phi_2(d, d; \rho^2) - p^2] / [p(1 - p)]$, where Φ_2 denotes the bivariate standard normal CDF and $d = \Phi^{-1}(p)$ is the default threshold. This equation is solved numerically for ρ using a Brent root-finding algorithm (`scipy.optimize.brentq`). Since both models are calibrated from the same default-rate data, their risk measures are structurally comparable.

An alternative approach to estimating the asset correlation would be to use the correlation of individual equity returns, provided the obligors are actively and liquidly traded. Düllmann, Küll, and Kunisch (2010) demonstrate in a comprehensive simulation study that equity-based estimators are more efficient, with their central result being that “the correlation estimators based on default rates are substantially downward biased in small samples.” Lucas (1995) had earlier pointed to the fundamental problems of historical default-correlation estimation: defaults are too rare and correlations too variable over time. Given the small portfolio sizes in the present dataset, the downward bias documented by Düllmann, Küll, and Kunisch (2010) is likely to affect the estimates presented here.

Despite the documented advantages of equity-based correlation estimation and the availability of the requisite data, this alternative is not pursued in the present analysis. The primary reason is methodological consistency: both models are calibrated from the same data source (historical default rates), ensuring that any differences in risk measures are attributable to model structure rather than input data.

3.2.1 Two-Factor Extension for Portfolio 3

Portfolio 3 differs structurally from the other two portfolios in that it combines obligors from both Industry A and Industry B. This raises the question of whether a single systematic factor adequately captures the dependence structure. Gordy (2003) notes explicitly that the single-risk-factor assumption is “much less innocuous from an empirical point-of-view” when the portfolio spans multiple industries with potentially distinct macroeconomic drivers.

Since no economic information beyond the industry labels is available, a formal assessment of whether the two industries are driven by different factors is not possible on substantive grounds alone. Nevertheless, a two-factor extension is implemented to empirically assess the sensitivity of the results to the factor structure.

In the extended model, two correlated factors F_A and F_B are simulated from a bivariate normal distribution. Each obligor is assigned to its respective industry factor. The inter-factor correlation is estimated from the daily returns of the two industry indices using the Pearson correlation coefficient, following the approach of Düllmann, Scheicher, and Schmieder (2007), who employ sector indices as proxies for latent systematic factors in a multi-factor credit portfolio model.

The simulation draws correlated factors via `numpy.random.multivariate_normal` with mean vector $(0, 0)$ and covariance matrix with ones on the diagonal and the estimated index correlation in the off-diagonal positions. The latent asset variable for each obligor is then:

$$X_i = \rho \cdot F_{\{Industry(i)\}} + \sqrt{(1 - \rho^2)} \cdot \varepsilon_i,$$

where ρ is the uniform factor loading from the one-factor model for Portfolio 3. An alternative would be to estimate industry-specific factor loadings; however, this would complicate the direct comparison with the one-factor specification without yielding additional insight given the data limitations.

4. Results and Discussion

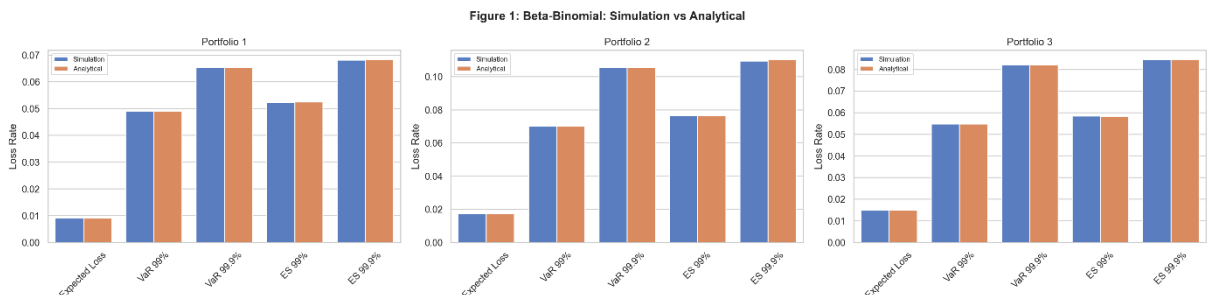
4.1 Beta-Binomial Model

Table 1 summarizes the key model parameters for all three portfolios. The estimated default correlations are small but uniformly positive. This finding is consistent with Zhou (2001), who establishes that default correlations are “generally very small over short horizons” and that “high credit quality implies a low default correlation over typical horizons.” The factor loadings ρ (approximately 9–21%) lie only partially within the 12–24% range prescribed by the Basel II IRB formula for corporate exposures (BCBS 2006).

Table 1: Beta-Binomial model parameters by portfolio

Portfolio	m	Total EAD	p	EAD-weighted LGD	Observed variance	Independent variance	ρ_y	s	α	β
Portfolio 1	28	631,058,707	1.99%	45.81%	0.00078608	0.00071566	0.37%	265.7617	5.2880	260.4737
Portfolio 2	28	704,450,945	3.52%	49.17%	0.00150996	0.00124632	0.81%	123.0921	4.3330	118.7591
Portfolio 3	35	769,806,387	3.12%	47.88%	0.00092319	0.00088199	0.14%	710.7209	22.1605	688.5604

The risk measures for all three portfolios are presented in Figure 1; exact values are reported in Table 3. Across all three portfolios, the simulated and analytical results are nearly identical, indicating that the Beta-Binomial distribution provides an adequate representation of the true loss distribution and that $N = 100,000$ scenarios are sufficient for stable empirical quantile estimation.



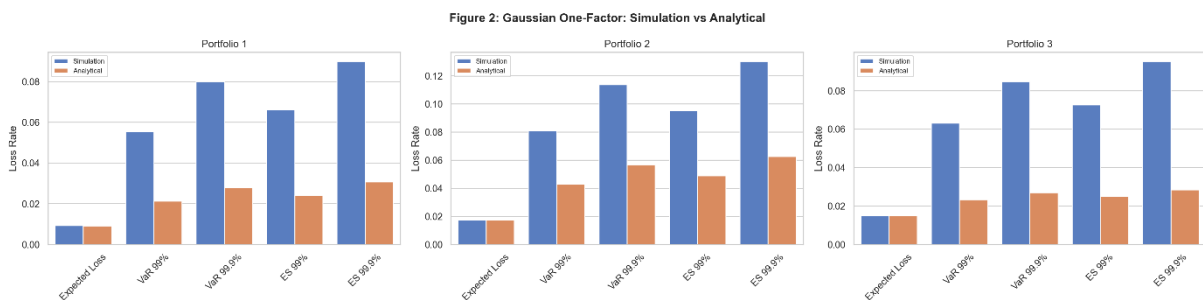
4.2 Gaussian Threshold Model

The asset correlations estimated for the one-factor model range from 0.85% to 4.24% (Table 2), substantially below the benchmarks reported in the literature and in the regulatory framework. The Basel II IRB formula prescribes asset correlations between 12% and 24% for corporate exposures (BCBS 2006), while Lopez (2004) estimates empirical values in the range of 10–30% based on equity returns. This finding is consistent with the property of the MM estimator documented by Düllmann, Küll, and Kunisch (2010), which systematically underestimates the true correlation in short time series.

Table 2: Gaussian one-factor threshold model parameters by portfolio

Portfolio	m	Total EAD	p	Threshold d	ρ_y	ρ	Asset corr. ρ^2
Portfolio 1	28	631,058,707	1.99%	-2.055868	0.37%	17.19%	2.96%
Portfolio 2	28	704,450,945	3.52%	-1.809311	0.81%	20.59%	4.24%
Portfolio 3	35	769,806,387	3.12%	-1.863723	0.14%	9.21%	0.85%

The most striking finding of the analysis is the pronounced discrepancy between the simulated and analytical risk measures (Figure 2). The Expected Loss is nearly identical across both evaluation methods for all three portfolios. The VaR and ES metrics, however, are systematically and substantially lower in the analytical computation across all portfolios and confidence levels. The root cause lies in the asymptotic nature of the analytical formula: Vasicek (2002) derives it explicitly for “very large portfolios,” a condition that none of the three portfolios (28–35 obligors) satisfies. In finite portfolios, idiosyncratic risk is not eliminated through diversification and remains present in the loss distribution. While the simulation correctly captures this residual risk, the analytical formula averages it away. Gordy (2003) formalizes this phenomenon as the “granularity bias” and demonstrates that “capital charges calibrated to the asymptotic case ... must understate required capital for any given finite portfolio.”

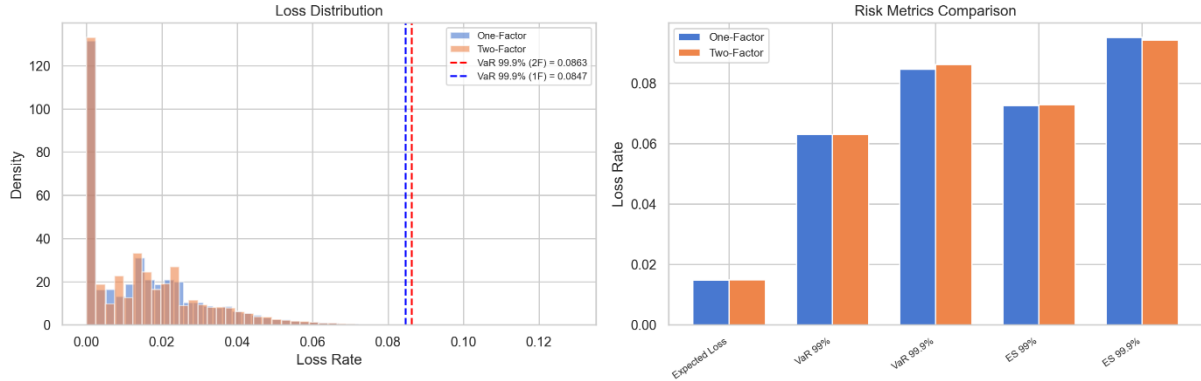


4.3 OneFactor vs. Two-Factor Gaussian Threshold Model

For Portfolio 3, a two-factor extension was developed that accounts for the distinct industry affiliations of the obligors and aims to capture the dependence structure more accurately. Figure 3 compares the simulated risk measures of the one-factor and two-factor specifications directly. The central observation is that both model variants yield nearly identical risk measures. This is initially surprising, since the introduction of a second factor with an inter-factor correlation of 0.458 would, in principle, be expected to produce a diversification benefit.

The explanation lies in the relatively small factor loading of only 9.21% for Portfolio 3. At this level, the idiosyncratic component dominates the asset return variance, leaving insufficient systematic exposure for the factor structure to materially affect the loss distribution. The limited discrepancy between the two specifications is therefore data-driven rather than a reflection of model equivalence.

Figure 3: Portfolio 3: One-Factor vs Two-Factor Gaussian Threshold



4.4 Comparison of the two Models

Table 3 presents the key risk measures of both models for all three portfolios. In the simulation-based evaluation, both models deliver comparable results, with the Gaussian one-factor model exhibiting systematically slightly higher tail risk measures. This finding is consistent with the analysis of Gordy (2000), who demonstrates the structural affinity of models in which default dependence is generated through common factors. Frey and McNeil (2003, Example 4.8) show explicitly that, given the same p and ρ_Y , different mixing distributions can yield similar but not identical tail measures.

Table 3: Final comparison: Beta-Binomial vs. Gaussian one-factor model

Model	Portfolio	Method	Expected Loss	VaR 99%	VaR 99.9%	ES 99%	ES 99.9%
Beta-Binomial	Portfolio 1	Simulation	0.91%	4.91%	6.54%	5.23%	6.81%
Beta-Binomial	Portfolio 1	Analytical	0.91%	4.91%	6.54%	5.25%	6.84%
Beta-Binomial	Portfolio 2	Simulation	1.72%	7.02%	10.54%	7.65%	10.96%
Beta-Binomial	Portfolio 2	Analytical	1.73%	7.02%	10.54%	7.66%	11.04%
Beta-Binomial	Portfolio 3	Simulation	1.49%	5.47%	8.21%	5.85%	8.45%
Beta-Binomial	Portfolio 3	Analytical	1.49%	5.47%	8.21%	5.83%	8.46%
Gaussian One-Factor	Portfolio 1	Simulation	0.92%	5.54%	7.99%	6.62%	8.99%
Gaussian One-Factor	Portfolio 1	Analytical	0.91%	2.13%	2.79%	2.41%	3.07%
Gaussian One-Factor	Portfolio 2	Simulation	1.73%	8.09%	11.37%	9.51%	13.02%
Gaussian One-Factor	Portfolio 2	Analytical	1.73%	4.28%	5.67%	4.88%	6.27%
Gaussian One-Factor	Portfolio 3	Simulation	1.49%	6.31%	8.47%	7.26%	9.52%
Gaussian One-Factor	Portfolio 3	Analytical	1.49%	2.34%	2.70%	2.50%	2.84%

The ratio of the 99.9% VaR to the EL reveals the influence of correlation more clearly. While Portfolio 1 has the lowest PD at 1.99%, it exhibits the largest dispersion of the loss distribution, with a VaR equal to 7.2 times the EL. Portfolio 3 presents the opposite picture: its PD of 3.12% is substantially higher, yet VaR and EL are closer together, indicating a less pronounced tail. The lower default correlation of Portfolio 3 leads to a relatively concentrated loss distribution,

whereas the higher correlation of Portfolio 1 generates a comparatively heavier tail despite the lower PD. This observation is consistent with Zhou (2001), who investigates the relationship between asset and default correlation and shows that the factor loading influences the shape of the loss distribution and, in particular, the probability of joint defaults.

5. Conclusions

The analysis yields four main findings. First, both models produce consistent and plausible risk estimates across all three portfolios in the simulation-based evaluation. Second, the analytical risk measures of the Gaussian one-factor model substantially understate the tail risk. This discrepancy is attributable to the granularity bias described by Gordy (2003): the analytical Vasicek formula is derived for asymptotically fine-grained portfolios, a condition that portfolios with 28 to 35 obligors do not satisfy. Third, the default correlations estimated from historical default rates are exceptionally low. The literature recommends equity-based estimation as a more reliable alternative (Düllmann, Küll, and Kunisch 2010), an approach that was not pursued in the present analysis in order to maintain methodological consistency across both models. Fourth, the two-factor extension for Portfolio 3, while theoretically well-motivated, yields only marginal changes in risk measures due to the small factor loading estimated from the data.

Taken together, the results underscore the necessity of simulation-based evaluation for small and concentrated portfolios, as well as the critical importance of the estimation methodology for the dependence parameters. A natural extension of the analysis would be to complement the default-rate-based correlation estimates with equity-based alternatives, as recommended by Düllmann, Küll, and Kunisch (2010), and to explore industry-specific factor loadings in the multi-factor framework.

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